

LIAM SHAW

liamshaw421@gmail.com · +91 93303 77645 · Kolkata, India · Open to relocation
github.com/Liam4Shaw · liamshaw.in

PROFILE

Computer Science graduate (FLAME University, 2026) with a Minor in Business Analytics and a full scholarship. Builds applied AI systems — multimodal deep learning, NLP, and full-stack GenAI products — end-to-end from architecture to deployed application. Also brings practical experience in CRM analytics, data cleaning, and campaign automation. Equally comfortable explaining model decisions to engineers and business insights to non-technical stakeholders. Open to roles across AI/ML engineering, data analytics, product analytics, and software engineering — and genuinely interested in growing toward product management and business-facing roles over time.

EDUCATION

FLAME University, Pune

2022 – 2026

B.Sc. (Hons) Computer Science · Minor: Business Analytics · Full Scholarship · School of Computing and Data Sciences

La Martiniere for Boys, Kolkata

2007 – 2022

ISC: 95% (2022) · ICSE: 92.4% (2020) · SAT: 1540 / 1600 (99th percentile)

TECHNICAL SKILLS

ML / AI: PyTorch · HuggingFace Transformers · scikit-learn · XGBoost · CNNs · BiLSTMs · NLP · RAG · NL2SQL · PCA · Random Forest · KMeans · DistilBERT · RoBERTa · ResNet-18

Languages: Python · Java · JavaScript · SQL · C · C++ · HTML/CSS

Frameworks & Backend: FastAPI · Flask · Spring Boot · REST APIs · JWT · SQLite · PostgreSQL · SQLAlchemy · Jinja · React · TypeScript · Node.js

Dev & Tooling: Docker · Maven · Git · GitHub · Jira · Jupyter · Bubble.io

Data & Analytics: CleverTap · pandas · NumPy · FuzzyWuzzy · Matplotlib · Excel · cohort analysis · funnel analysis · campaign automation

PROJECTS

MoodTrack — Multimodal & Multilingual Music Emotion Recognition · Graduation Project · Supervisor: Prof. Manoranjan Dash

Mar 2026

- Addressed a fundamental gap in Music Emotion Recognition: existing systems reduce a full track to a single static label, discarding temporal dynamics. MoodTrack produces continuous arousal and valence predictions at 0.5-second resolution across 1,802 annotated songs (DEAM dataset) and 277 individual listener profiles across 54 tracks (MERP dataset).
- Architecture:** raw MP3 → Mel-spectrogram (128 Mel bins, 22.05 kHz, librosa) → CNN encoder (3 × Conv2D-BatchNorm-ReLU-MaxPool, global average pooling) → two-layer BiLSTM (hidden 64, bidirectional) → per-segment arousal and valence regression heads clipped via tanh to $[-1, 1]$.
- Audio-only baseline:** arousal CCC = 0.738 (within 5.4% of Chaki et al. 2020 benchmark: 0.780). Early-fusion variants tested: simple concatenation and cross-modal attention (V2); V2 degraded to 0.690 due to all-zero lyric vectors on 99.3% of training tracks — documented as expected behaviour with sparse lyric coverage.
- Late-fusion dual-tower model** (best result): arousal CCC = 0.774, valence CCC = 0.441. Learned scalar gates converged to $\alpha = 0.77$ (audio-dominant for arousal) and $\beta = 0.36$ (lyric-dominant for valence) — independently recovering an established music-psychology finding without any explicit prior.
- Personalised emotion sub-model** (MERP): tri-branch architecture (audio + lyrics + ProfileMLP) conditioned on 12-dimensional demographic vectors; PersonalisationAttention reweights audio features by listener profile. Arousal CCC = 0.533 across 5 listener archetypes (General, Young Pop Fan, Trained Musician, Older Classical, Metal Fan) — demonstrating measurably distinct perceived-emotion trajectories by demographic group.
- MoodTrack web application:** Flask backend, vanilla JS + Chart.js frontend; real-time MP3 upload, dual-line trajectory visualisation with ground-truth overlay, persona-aware recommendation engine (weighted Euclidean distance, valence 1.5× weighted) over a pre-computed emotion embedding cache.

ScoutAI — AI-Powered Football Scouting & Analytics Platform · Applied Generative AI · FLAME University

Apr 2026

- Full-stack generative AI application for professional football recruitment. Backend: FastAPI + SQLite (~370 MB, 14 ORM-mapped tables via SQLAlchemy) + JWT/bcrypt authentication. Frontend: 8-module Single Page Application.
- **NL2SQL pipeline**: user natural language → schema-prompted LLM → error fixing → SELECT-only validation → execution → natural-language insight generation. Primary LLM: Groq llama-3.1-8b-instant; failover: Ollama/mistral for local deployment. Temperature = 0 for deterministic outputs.
- **ML analytics subsystem**: PCA (2D player space projection), Random Forest (archetype classification, 17-metric feature vector, global feature importance + z-scores), KMeans offline archetype clustering, cosine-similarity nearest-neighbour player search, RAG-enhanced LLM inference for statistical summaries.
- **Agentic features**: (1) scout-note entity extraction with async LLM background processing; (2) profile-contextualised prompt injection; (3) autonomous transfer-recommendation agent orchestrating scout search, similarity, and report tools in sequence; (4) JWT-bound cross-request identity resolution.
- Chain-of-thought scouting pipeline: LLM independently identifies statistical strengths, weaknesses, and synthesises a 200–300 word professional verdict. 20+ REST endpoints; complete ETL pipeline; all four agentic initiatives delivered and integrated.

Hybrid Fake News Detection (AMR Semantic Features + RoBERTa) · NLP Research · FLAME University Dec 2025

- Hybrid text classifier combining RoBERTa contextual embeddings (768-dim) with 26 hand-engineered AMR structural features: entity density and coreference (6), syntactic passivisation and tree depth (8), semantic modality markers (7), sentiment and style features (5).
- **87.07% accuracy** on FakeNewsNet (vs. 60–70% for standard ML and pure-transformer baselines). Focal loss addressed class imbalance. Threshold optimisation (0.5 → 0.1) independently contributed ~3% gain.
- Ablation study: RoBERTa alone reached ~69%; combined model reached 87.07%. Feature importance analysis identified passivisation patterns and entity-type distributions as the top AMR discriminators for misinformation detection.

Multimodal Music Emotion Recognition (Cross-Attention Fusion) · NLP Course Project · FLAME University Dec 2025

- Team project: ResNet-18 audio encoder (Mel-spectrogram, grayscale-adapted pretrained weights) + DistilRoBERTa text encoder (BPE tokenisation, 256-token context) fused via learned Gated Attention (dynamic $\alpha \in [0, 1]$) on MERGE-Balanced (2,000 tracks, 4 balanced quadrants).
- **78% test accuracy** (300-song held-out set); F1 = 0.91 for Angry/Tense quadrant. SpecAugment data augmentation reduced train-validation gap by ~18pp. Grad-CAM confirmed the model learned musically meaningful spectral features.

FLAME Moodle Insight Dashboard · Software Architecture & Engineering (CSIT306) · Team Project 2025

- **React + TypeScript + Tailwind CSS** analytics dashboard for FLAME University's Center for Digital Learning — replacing a manual, hours-long Moodle CSV-to-Excel student engagement workflow with a local-first, privacy-preserving dashboard (no backend, no data leaves the device, no IT access required).
- Weighted engagement scoring algorithm with a **traffic-light flagging system** that surfaces at-risk students automatically; scoring weights are tunable by CDL admins without code changes.
- Four sit-down client consultation rounds with the CDL team; features were iterated based on actual workflow — 'we just want to see who's red' directly shaped the final interface. Deployed and in active use for pre-orientation cohort tracking.

EXPERIENCE

ACEplus (Derek O'Brien & Associates · Edtalk Edtech Pvt. Ltd.) · Analytics & CRM Intern · Kolkata May – Jul 2025 · Hybrid

- Cross-functional role embedded in the School Relationships team, collaborating with an outsourced dev team in Gurugram on analytics implementation, CRM configuration, and feature testing.
- **Analytics audit**: Identified 50+ inconsistently named CleverTap events; designed Zone_Module_Action naming convention (e.g., LZ_Beginner_Capsule1_Completed) and a 3-phase migration plan (legacy → parallel fire → CleverTap backfill) — formally adopted by the engineering team.
- **Event specification**: Authored a 50+ event proposal documenting event names, trigger points, property types, value ranges, and use-case campaigns. Delivered a companion Activity List ensuring no in-app interaction was left untracked across Learning, Gaming, and Registration zones.
- **Cohort insight**: Workshop-driven user cohort earned ~2× more in-app ACEs than organic signups — finding adopted by the product team to prioritise school engagement over organic acquisition.

- **Data cleanup:** Python + FuzzyWuzzy batch script resolved ~15% of blank/inconsistent school-name profile fields, improving segmentation accuracy for school-specific push campaigns and weekly coach reports.
- Built user segments and audience filters in CleverTap across demographic, behavioural, and engagement dimensions; configured and tested retention/re-engagement automation journeys (push, in-app, email, WhatsApp); tested offline event-caching behaviour and validated new feature integrations with the dev team via Jira.

Mandrake Tech · Software Intern · Pune

Oct 2024 – Jan 2025

- Built a containerised URL-shortening service — **Spring Boot, PostgreSQL, JWT authentication** — with Docker + Maven configuration that removed environment-parity issues across the development team.
- Implemented REST-integrated frontend interfaces; owned the full request lifecycle from client UI through API routing to database persistence.

Willwali · Software Developer Intern · Remote · Kuala Lumpur

May – Jul 2024

- Part of the first intern cohort at an early-stage Malaysian legal-tech startup automating will-writing across multiple legal jurisdictions. Stack: Python · Flask · Jinja · Bubble.io · Git.
- Developed CRUD microservices in Flask/Jinja; wrote **50+ unit and end-to-end tests (90%+ coverage)** with unittest and pytest, covering sorting, HTTP endpoints, error handling, and routing.
- Generated test wills across 5+ demographic and jurisdictional profiles; identified and documented **30+ bugs** in the Bubble MVP spanning Hindu Succession Act, Islamic Faraid distribution, and Malaysian inheritance law compliance.
- Established team Git workflows (branching, PRs); maintained PEP 8 style, docstrings, function annotations, and README documentation throughout.

CERTIFICATIONS

CS50x: Introduction to Computer Science · HarvardX / edX

Building AI Literacy · LinkedIn Learning · Mar 2025

Responsive Web Design · freeCodeCamp

ACTIVITIES

Sports Analytics & Reporting — FLAME University: Consolidated and analysed club activity and media performance datasets across sports verticals; derived insights across clubs and audience segments; produced final analytical reports (18.91M total views, 6.57M accounts reached, 79,100+ followers gained, 72.4% non-follower view share).